

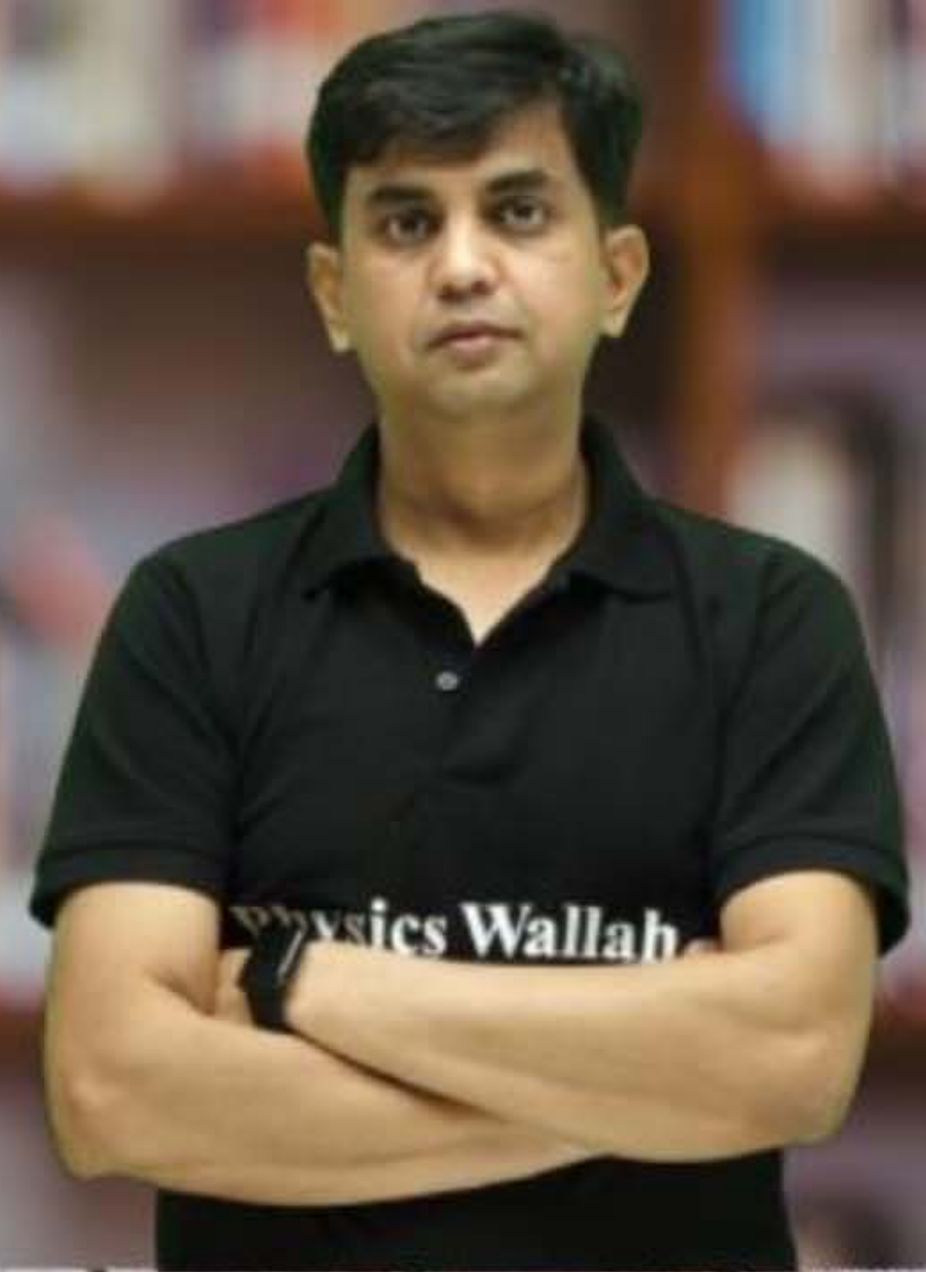
CS & IT ENGINEERING



Computer Network

MAC Sub-Layer

Lecture No. - 02



By - Abhishek Sir



Recap of Previous Lecture



Topic

MAC Sub-Layer

Topic

Multiple Access Protocol



Topics to be Covered



Topic

Pure ALOHA

Topic

Slotted ALOHA

ABOUT ME



Hello, I'm **Abhishek**

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Topic : Pure ALOHA



- > When node has packet to transmit, it transmit immediately
- > Allow collision to happen
[recover via "retransmission"]
- > Use randomization in choosing "when to retransmit"
[Back-off time]

Not immediately
after some random wait time.

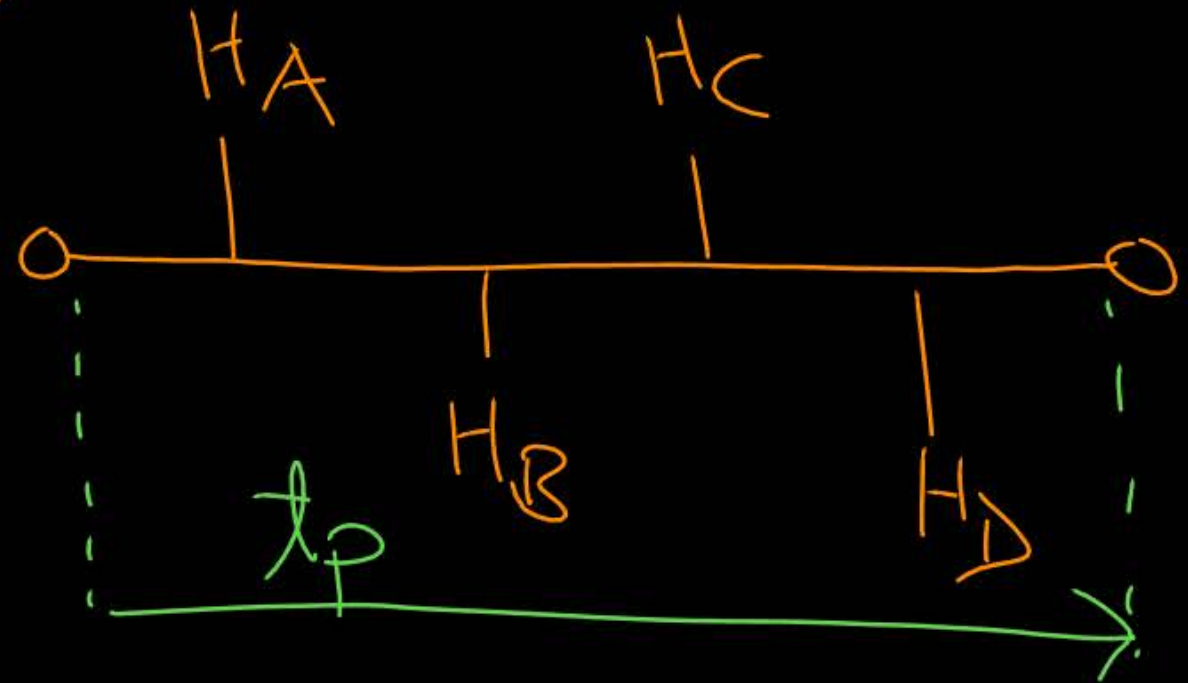


Topic : Pure ALOHA



- > All frames are of equal size
- > Receiver send acknowledgement to Source
[for every correctly (without collision) received frame]
- > After transmitting a frame, Source wait for an ACK upto time-out
- > Time-out time = Round Trip Propagation Delay

$$= 2t_p$$





Topic : Pure ALOHA





Topic : Pure ALOHA



Vulnerable time :

-> Time duration in which no transmission should be done to avoid collision

↓
start

Vulnerable time = 2 * Frame Transmission Time

$$= 2 * t_x$$



Topic : Pure ALOHA



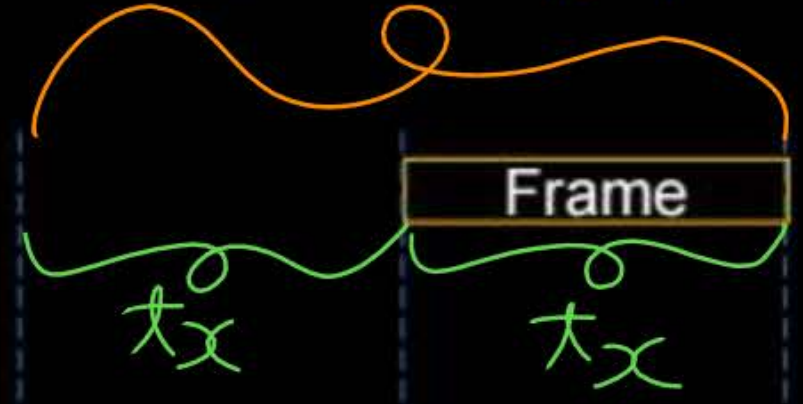
A

B

C

D

Vulnerable time for this frame



$t - t_x$

t

$t + t_x$

Time

Example 1 :-

Consider a network using the pure ALOHA medium access control protocol,
where each frame is of length 2,000 bytes. The channel transmission rate is 1
Mbps. Calculate vulnerable time (in milliseconds) ?

Solution :

$$\text{Frame Size} = 2000 \text{ bytes} = 16 * 10^3 \text{ bits}$$

$$\text{Bandwidth} = 1 \text{ Mbps} = 10^6 \text{ bits / sec}$$

$$t_x = \frac{\text{Frame Size}}{\text{Bandwidth}} = \frac{16 * 10^3 \text{ bits}}{10^6 \text{ bits / sec}} = 16 * 10^{-3} \text{ sec} = 16 \text{ ms}$$

For Pure ALOHA :

$$\begin{aligned} \text{Vulnerable time} &= 2 * \text{Frame Transmission Time} \\ &= 2 * t_x = 32 \text{ ms} \end{aligned}$$

$$\boxed{\text{Ans} = 32}$$



Topic : Pure ALOHA



G : Mean number of transmissions per frame time

[Average number of frame transmitted in one frame transmission time]

S : Throughput per frame time

[Average number of frame transmitted successfully in one frame transmission time]

P = Probability that a frame does not involve in a collision

$$S = G * P$$

$$S < G$$

Example 2 :-

Consider a network using the pure ALOHA medium access control protocol, where each frame transmission delay is 10 milisecond. The aggregate number of transmissions across all the nodes is 5,000 frames per second. Calculate mean number of frame transmissions per frame time?

$$t_x = 10 \text{ ms}$$

$$G = ?$$

$$1 \text{ sec} \longrightarrow 5000 \text{ frames}$$

$$10^3 \text{ ms} \longrightarrow 5000 \text{ frames}$$

$$1 \text{ ms} \longrightarrow 5 \text{ frames}$$

$$(t_x = 10 \text{ ms}) \longrightarrow 50 \text{ frame}$$

$$G = 50$$

$$\text{Ans} = 50$$



Topic : Pure ALOHA



According to Poisson's distribution

$$P = e^{-2G}$$

then

$$S = G * e^{-2G}$$



Example 3 :-

Consider a network using the pure ALOHA medium access control protocol, where each frame transmission delay is 2 milisecond. The aggregate number of transmissions across all the nodes is 4,000 frames per second. Calculate throughput per frame time?

$$t_x = 2 \text{ ms}$$

$$1 \text{ sec} \longrightarrow 4000 \text{ frames}$$

$$10^3 \text{ ms} \longrightarrow 4 \times 10^3 \text{ frames}$$

$$1 \text{ ms} \longrightarrow 4 \text{ frame}$$

$$2 \text{ ms} \longrightarrow 8 \text{ frame}$$

$$G = 8$$

Pure ALOHA :-

$$S = G * e^{-2G}$$

$$= 8 * e^{-16}$$

$$= 9.0028 * 10^{-7}$$

$$\approx 9.45 * 10^{-7}$$

#Q. Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 1 Mbps ($=10^6$ bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is _____.

[GATE 2021]

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Topic : Pure ALOHA



$$S = G * e^{-2G}$$

For what value of G throughput S is maximum ?

$$\left[\frac{dS}{dG} = 0 \right]$$

$$S = G \times e^{-2G}$$

$$\frac{dS}{dG} = \frac{d}{dG} [G \times e^{-2G}]$$

$$= G \times \frac{d}{dG} e^{-2G} + e^{-2G} \times \frac{d}{dG} G$$

$$= (G \times e^{-2G} \times -2) + e^{-2G}$$

$$= (e^{-2G} - 2G e^{-2G})$$

$$= e^{-2G} (1 - 2G)$$

$$\frac{dS}{dG} = 0$$

$$e^{-2G} (1 - 2G) = 0$$

$$1 - 2G = 0$$

$$G = \frac{1}{2}$$

$$S = G \times e^{-2G}$$

at $G = \frac{1}{2}$, max^m throughput per frame time

$$S = \frac{1}{2} \times e^{-2 \times \frac{1}{2}}$$

$$[e = 2.718281828]$$

$$S = \frac{1}{2} \times e^{-1}$$

$$S = \frac{1}{2e}$$



Topic : Pure ALOHA



$$S = G * e^{-2G}$$

Maximum throughput can achieve at $G = 0.5$

$$\text{Maximum throughput} = 1/2e = 0.1839 = 18.39\%$$



2 mins Summary



Topic

Pure ALOHA ✓

Topic

~~Slotted ALOHA~~



THANK - YOU

